Learning from Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Learning from Examples

> Prepare Learning from Examples: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Debug the Cause

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Look for: Compare expected and observed traces to locate the first difference

Reference example, not the only acceptable wording: Expected trace: 3, 5, 10, 9. First difference is the doubling checkpoint: 10 expected versus 8 observed.

The later 8 minus 1 equals 7, so the observed subtraction is consistent with its **input**. Investigate the earlier step before claiming another fault.

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate a condition using the current state and trace only the selected branch

Reference example, not the only acceptable wording: Start 2: after adding 2 score is 4; at least 6 is false; add 4 and report 8. Start 5: after adding 2 score is 7; at least 6 is true; subtract 3 and report 4.

Each run starts fresh and executes only its selected branch.

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a category definition using explicit **evidence** and boundary rules

Reference example, not the only acceptable wording: In order: ready because both known fields pass; revise because 31 exceeds 30; review because the known day passes but the room is unknown; revise because the known day failure has priority over the unknown room.

Equality at 30 is accepted.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit **training**.

Choose one new labelled card length that would distinguish them and explain what remains unknown.

Concept to explain

Training examples influence how a model handles later **input**s.

Term	Meaning
training example	input paired with a target label
pattern	recurring relationship across examples
training	fitting behavior from examples

Term	Meaning
new input	case not used in fitting

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Extend Week 4 from a small candidate comparison to an explicit **fitting** algorithm, recorded tie handling and a changed **fitting set**. Use the same examples for every candidate and reset counts.

The chart deliberately includes zero errors and a tie. Do not present the supplied classroom labels as natural truth or the selected threshold as a discovered universal law.

Readiness answer

Predictions are short, tall and tall. At 5 the at least rule includes equality; an accidental greater than rule would instead predict short.

Four tower cards showing height on the front and supplied label on the back

Candidate cards 3, 5, 6 and 7, a separate tally for each, and the smaller-threshold **tie rule**

An additional 5 cm short card, the independent distance examples and two conflicting 6 cm records

End goal for the pictorial case

By the end, I can compare **candidate model**s, resolve a tie explicitly and separate fit from **prediction**.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Learning from Examples using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT labeled example cards

2. Process — the deciding step

PROCESS adjust a decision **pattern** from recurring clues

3. Output — the result

OUTPUT prediction for a new card

Key idea

CORE IDEA **Training example**s influence how a model handles later **input**s.

How the mechanism works

UNDER THE HOOD Supervised learning uses labeled examples to fit a mapping from represented **input**s to **output**s.

The model does not store a human rule automatically; its fitted behavior reflects correlations, coverage, labels, and optimization choices.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A learner shown mostly striped triangles may wrongly treat stripes as the important clue for the triangle label.

Step 1

STEP 1 — OBSERVE labeled example cards

Step 2

STEP 2 — TRACE adjust a decision **pattern** from recurring clues

Step 3

STEP 3 — CONCLUDE **prediction** for a new card

Model answer

MODEL REASONING Early examples may support several rules. New examples eliminate some possibilities and may change the decision boundary. Record each revision as **evidence** that **training** data shapes later behavior.

A claim to avoid

DO NOT OVERCLAIM Learning from examples is not the same as memorizing every example or understanding the concept like a person.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

The teaching case

Predict and state the **evidence** limit.

Exact explanation to model

Why the result follows

Predict T from the demonstrated edge **pattern**. One success does not establish broad reliability; use fresh labelled tests including ambiguous cases.

Explain the added worked example

Four imaginary paper towers have heights 2, 4, 6 and 8 centimetres. Their supplied labels are short, short, tall and tall. These labels belong to this classroom task; they are not a universal definition of tall.

We will fit a rule using height as its only **input**.

Explain the added worked example · 1

Every candidate predicts tall when height is at least its threshold and short otherwise. Compare thresholds 3, 5, 6 and 7 centimetres. The **fitting** procedure chooses the fewest disagreements with the four labels.

If candidates tie, choose the smaller threshold. Write these choices before comparing results.

Explain the added worked example · 2

For each candidate, start its count at zero. For each example, predict with that candidate, compare with the supplied label, and add one only for a disagreement. After all four examples, record the count.

Then move to the next candidate with a fresh count. This uses the conditions and repetition you have already traced.

Threshold 3 incorrectly calls the height 4 tower tall. Threshold 7 incorrectly calls the height 6 tower short. Thresholds 5 and 6 agree with all four supplied labels.

Check each disagreement against the individual example instead of trusting a total without its trace.

The bar chart compares **disagreement count**s, so a shorter bar is better on this **fitting set**. Zero is a measured count here, not missing data. This chart says nothing yet about the error count on new towers.

Key terms

Term	Meaning
Candidate model	One possible rule being compared during fitting
Fitting set	The examples used to choose or adjust a model

Term	Meaning
Disagreement count	The number of predictions that differ from their comparison labels
Tie rule	An explicit method for choosing between equally scoring candidates
Generalisation	How well a fitted model works on new cases beyond its fitting examples

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Height and supplied label	Prediction under threshold 5	Prediction under threshold 6
2 cm short	Short	Short
4 cm short	Short	Short
6 cm tall	Tall	Tall
8 cm tall	Tall	Tall
5 cm label not yet supplied	Tall	Short

Model the reasoning aloud · 3

The first four rows are **fitting** examples. The final row asks what the two tied candidates predict at a height missing from that set. Our **tie rule** selects 5, so the selected model predicts tall at 5.

The **fitting** examples alone do not tell us whether that new **prediction** matches the task label.

A **tie rule** makes the selection reproducible. It does not prove that the chosen threshold is the true boundary. Record that 6 tied with 5 and that they disagree at height 5.

Hiding the tie would hide a real limitation of the available **evidence**.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Learning from Examples

A fictional classifier learns whether measured sticks are short or long.

The facts we will use

Supplied facts

Training lengths 2,4,8,10 cm have labels short,short,long,long.

Supplied facts

Candidate boundaries: long at least 5 or at least 7. Lowest errors wins; tie chooses smaller boundary.

AI ACADEMY / AI-1/13

Same training errors, different new prediction

	Boundary	Training errors	Prediction at 6
Candidate 5	5	0	long
Candidate 7	7	0	short

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/13

Learning from Examples

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training rows

2 and 4 are short; 8 and 10 are long.

2. Candidate checks

Boundaries 5 and 7 both predict S,S,L,L: zero errors each.



3. Tie policy

Choose boundary 5 because the declared tie rule prefers the smaller value.



4. New prediction

Length 6 is predicted long by the selected rule.

CHANGE ONE THING

Reverse the tie policy to prefer the larger boundary.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Training rows	2 and 4 are short; 8 and 10 are long.
2. Candidate checks	Boundaries 5 and 7 both predict S,S,L,L: zero errors each.
3. Tie policy	Choose boundary 5 because the declared tie rule prefers the smaller value.
4. New prediction	Length 6 is predicted long by the selected rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

There is no **training** stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true.

Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label.

What this example does not establish

Keep the stated limit

Zero **training** errors for both rules cannot determine the correct label of every unobserved stick.

Change one thing in the pictured case

Change one thing

Change: Reverse the tie policy to prefer the larger boundary.

Predicted result

Prediction: Boundary 7 is selected and predicts short at 6.

Why the result follows

Reason: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Learning from Examples, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Training example**s label smooth leaves S and toothed leaves T. A new leaf is toothed.

2. State the decision to explain

Write the question to explain: Predict and state the **evidence** limit.

3. Reveal the worked reasoning

Reveal after **prediction**s: Predict T from the demonstrated edge **pattern**. One success does not establish broad reliability; use fresh labelled tests including ambiguous cases.

Hold back the independent task

Keep for an independent attempt: A learner sees three red apples and no green apples, then calls a green apple not an apple. What missing **evidence** could help? How would you check improvement?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

At height 5, threshold 5 predicts tall and adds one disagreement; its total is 1. Threshold 6 predicts short and adds none; its total remains 0. Select 6 between these candidates.

The new label affected selection and is therefore **fitting evidence**. Exit: the threshold agrees with these four supplied labels under this procedure, but future performance remains unestablished.

Check **prediction**s on appropriate new labelled examples kept separate from selection and inspect their coverage.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve at least and its equality case. Ask what information is available when predicting a new label.

Stage	Teacher action
7 to 20 minutes	Model threshold 3 with a fresh zero tally. Predict each example before comparing its label. Record the single disagreement at height 4.
20 to 33 minutes	Complete all four candidates. Interpret the shared-scale chart and emphasise that zero-length bars represent zero disagreements.
33 to 45 minutes	Complete W1 and W2 in pairs. Apply the tie rule but preserve the record of both tied candidates and their prediction difference at height 5.
Session 2 start 0 to 8 minutes	Retrieve the selected threshold and ask whether the tie rule proves its missing boundary label.
8 to 23 minutes	Use the guided task and W3 to add height 5 short. Recompute on all five examples and explicitly reclassify the added evidence as fitting data.
23 to 38 minutes	Collect W4 independently. Require all prediction lists and counts, then cover the new label while students predict at distance 2.
38 to 45 minutes	Discuss W5 and the exit. Separate record investigation from changing labels to flatter the model. Preview constructing a complete classifier.

Reduce help deliberately

Completion practice

Completion task: Thresholds 5 and 6 both have zero disagreements on the W1 **fitting set**. The **tie rule** chooses the smaller threshold. Which is selected?

Give a whole-number height where the two candidates disagree and both **prediction**s. Explain what the **tie rule** does not prove.

Reasoning to listen for

Teacher check: Select 5. At height 5 cm, threshold 5 predicts tall and threshold 6 predicts short.

Choosing the smaller threshold makes the procedure definite; it does not prove which **prediction** matches a label not supplied by the **fitting set**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Fitting** a **pattern** from examples. Proposed explanation: “The lowest error count is added across candidates”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Reset to zero for each candidate so all totals describe the same examples under one rule.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Which new measured length would reveal disagreement between the candidates?

Reasoned teacher answer

Length 6 does: boundary 5 predicts long and boundary 7 predicts short. A checked label would then provide new **evidence**.

Changed-case reasoning: Boundary 7 is selected and predicts short at 6. The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Learning from Examples investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Act as a model trained on shape cards; record rule changes.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR Remembering teaching examples proves learning works everywhere.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The lowest error count is added across candidates: Reset to zero for each candidate so all totals describe the same examples under one rule.

A **tie rule** proves the selected boundary is correct: Compare tied thresholds at height 5, which was absent from the original **fitting set**.

Zero **fitting** errors guarantee future correctness: Separate the **evidence** used to choose the rule from new labelled cases used to evaluate it.

Disagreeing labels should be changed to match the model: Inspect original measurements, units, omitted information and label definitions; model agreement is not sufficient grounds for editing a record.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Learning from Examples questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Predict T from the demonstrated edge **pattern**. One success does not establish broad reliability; use fresh labelled tests including ambiguous cases.

Independent question

Independent question · withhold the key

A learner sees three red apples and no green apples, then calls a green apple not an apple. What missing **evidence** could help? How would you check improvement?

Expected independent answer

Teacher key · Expected independent answer

Add varied labelled apple examples, including green apples. Check with separate new examples not used to teach the learner. More examples help only if they provide useful, correct **evidence**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Remembering teaching examples proves learning works everywhere.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> A card of length 5 or 6 receives L under the first rule and S under the second. Its independently established label helps compare the rules.
>
> **Training** fit alone does not determine a unique rule; one extra label still does not settle all future cases.

Explanations for the additional workbook examples

W1 Count each candidate from zero

Workbook question · W1

Heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Predict all four labels under threshold 3, then under threshold 7, using tall when height is at least the threshold.

List the disagreeing examples and separate totals.

Teacher reasoning · W1

Threshold 3 predicts short, tall, tall, tall, disagreeing only at 4 cm: count 1. Threshold 7 predicts short, short, short, tall, disagreeing only at 6 cm: count 1. Reset the tally for each candidate.

W2 Explain a tied fit

Workbook question · W2

Thresholds 5 and 6 both have zero disagreements on the W1 **fitting set**. The **tie rule** chooses the smaller threshold. Which is selected? Give a whole-number height where the two candidates disagree and both **prediction**s.

Explain what the **tie rule** does not prove.

Teacher reasoning · W2

Select 5. At height 5 cm, threshold 5 predicts tall and threshold 6 predicts short.

Choosing the smaller threshold makes the procedure definite; it does not prove which **prediction** matches a label not supplied by the **fitting set**.

W3 Update with an added example

Workbook question · W3

Add height 5 cm labelled short to the four **fitting** examples. Recompute the disagreement totals for thresholds 3, 5, 6 and 7. Select the candidate with the fewest disagreements, keeping the smaller-threshold **tie rule**.

Explain why the new label is part of **fitting**.

Teacher reasoning · W3

Totals are 2, 1, 0 and 1 respectively. Threshold 3 now disagrees at 4 and 5; threshold 5 at 5; threshold 6 nowhere; threshold 7 at 6. Select 6. The new label influenced selection, so it is **fitting evidence**.

W4 Fit a new distance task

Workbook question · W4

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units.

List every **prediction** and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

Teacher reasoning · W4

Threshold 2 predicts near, far, far, far with 0 errors. Threshold 4 predicts near, near, far, far with 1 error at 3. Threshold 6 predicts near, near, near, far with 2 errors at 3 and 5. Select 2 and predict far at exactly 2.

This fits the four labels among these candidates; it does not establish the unknown label at 2 or all future distances.

W5 Keep conflicting records visible

Workbook question · W5

Two **fitting** records both have height 6 cm. One is labelled short and the other tall. A learner says a clever height-only rule can agree with both on every run.

Explain the problem for a deterministic rule and name two checks to make before changing either record.

Teacher reasoning · W5

The same **input** must receive the same **prediction** from this deterministic height-only rule, so at least one of the two labels disagrees.

Check the original measurements and units, and check whether both labels used the same task policy; also investigate an omitted relevant feature. Do not edit a label merely to improve the score.

Assess independent transfer

Independent question · withhold the key

Selected case: Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units.

List every **prediction** and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

Selected case answer

> **Teacher key · Selected case answer**
>
> Threshold 2 predicts near, far, far, far with 0 errors. Threshold 4 predicts near, near, far, far with 1 error at 3. Threshold 6 predicts near, near, near, far with 2 errors at 3 and 5. Select 2 and predict far at exactly 2.
>
> This fits the four labels among these candidates; it does not establish the unknown label at 2 or all future distances.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Learning from Examples to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Create the first small labeled example set for the sorter and explain which visual differences it covers.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 14: Build a Human Classifier. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use counters for disagreements and a number line with a closed threshold dot. Let students read labels aloud after committing **prediction**s.

Extension: identify all positive whole-number thresholds with zero disagreements on the original tower set and explain why the available labels do not select uniquely between them.

W4 is secure when the learner lists the three **prediction** sequences, obtains counts 0, 1, 2, selects 2, includes equality in the new **prediction** and limits the claim.

After support, retry values 2, 4, 6, 8 labelled near, near, far, far with thresholds 4 and 6: counts 1 and 0, select 6. Ask about an unseen distance 5 to check that **prediction** is not confused with verified correctness.

Week 14 constructs a classifier as a complete procedure with defined **input**s, a selected model and explicit handling of invalid or missing records.

Plan the delayed return

Suggested return points in this level: ai-1/14, ai-1/16, ai-1/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Save candidate definitions, **training** errors and tie policy with any sorter model version.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Learning from Examples lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Thresholds 5 and 6 both have zero disagreements on the W1 **fitting set**. The **tie rule** chooses the smaller threshold. Which is selected?

Give a whole-number height where the two candidates disagree and both **prediction**s. Explain what the **tie rule** does not prove.

Supported practice answer

Reasoned practice answer: Select 5. At height 5 cm, threshold 5 predicts tall and threshold 6 predicts short.

Choosing the smaller threshold makes the procedure definite; it does not prove which **prediction** matches a label not supplied by the **fitting set**.

Check every learner before discussion: Use the worked case for **Fitting** a **pattern** from examples. Proposed explanation: “The lowest error count is added across candidates”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Reset to zero for each candidate so all totals describe the same examples under one rule.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Reverse the tie policy to prefer the larger boundary. Require a **prediction** and a reason before comparing. Expected result: Boundary 7 is selected and predicts short at 6.

Reason: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

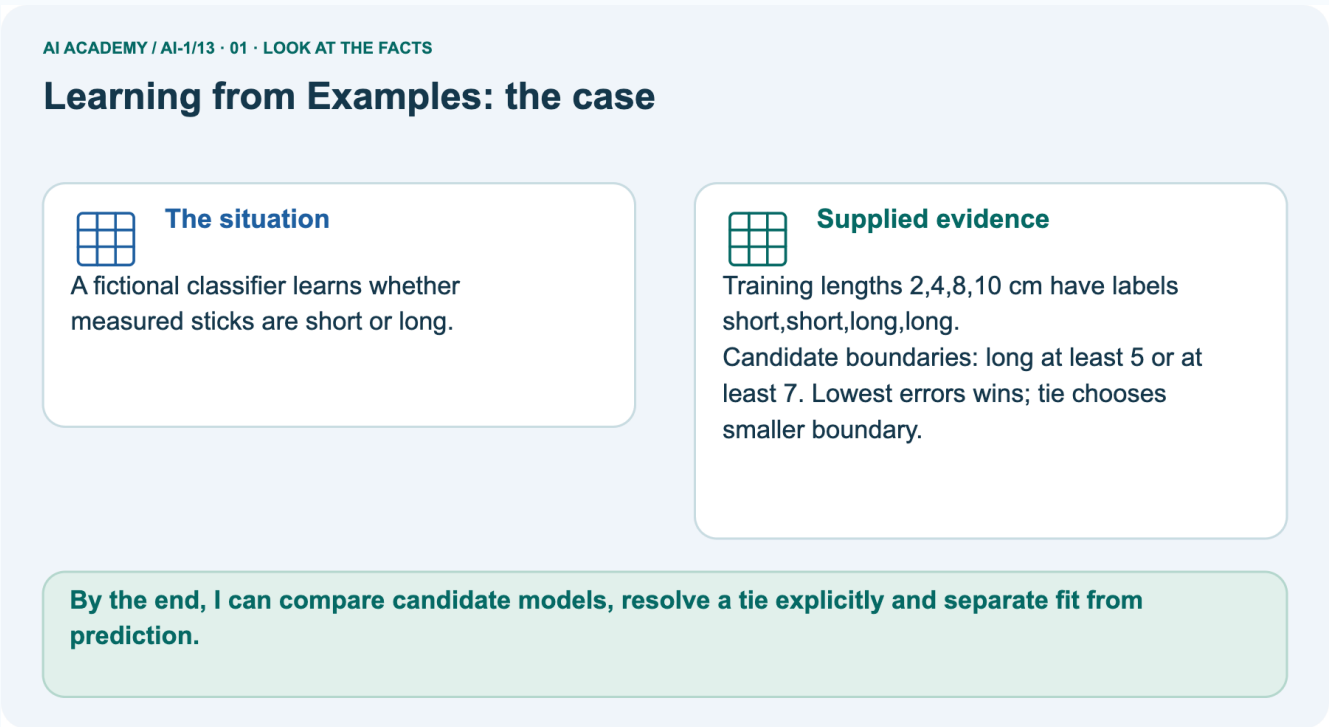
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Learning from Examples graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Learning from Examples: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classifier learns whether measured sticks are short or long. Supplied facts: **Training** lengths 2,4,8,10 cm have labels short,short,long,long. Candidate boundaries: long at least 5 or at least 7.

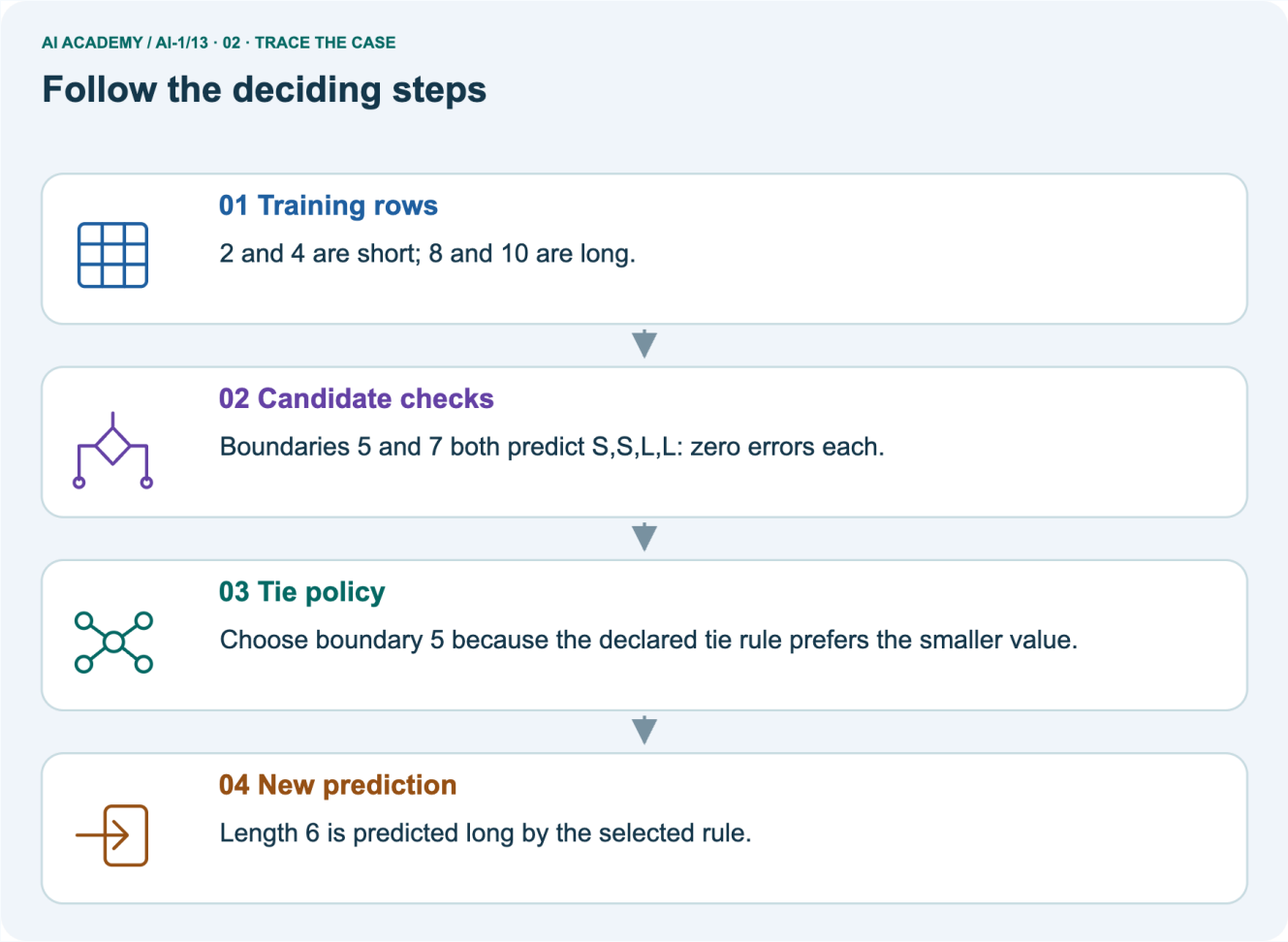
Lowest errors wins; tie chooses smaller boundary.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

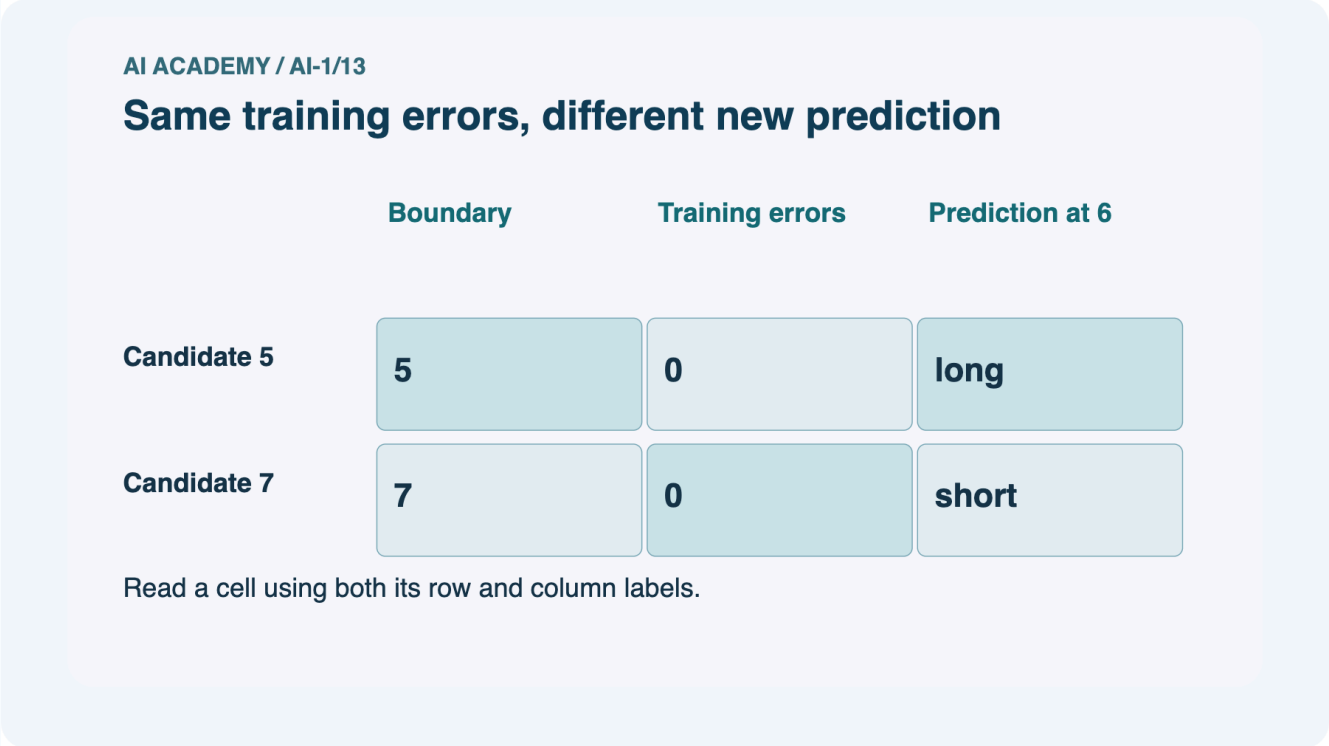
Training rows: 2 and 4 are short; 8 and 10 are long. → Candidate checks: Boundaries 5 and 7 both predict S,S,L,L: zero errors each. → Tie policy: Choose boundary 5 because the declared **tie rule** prefers the smaller value. → New **prediction**: Length 6 is predicted long by the selected rule.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Same training errors, different new prediction



Follow the labelled picture with the matching explanation. Open the vector diagram

Same **training** errors, different new **prediction**. Candidate 5: Boundary = 5, **Training** errors = 0, **Prediction** at 6 = long. Candidate 7: Boundary = 7, **Training** errors = 0, **Prediction** at 6 = short.

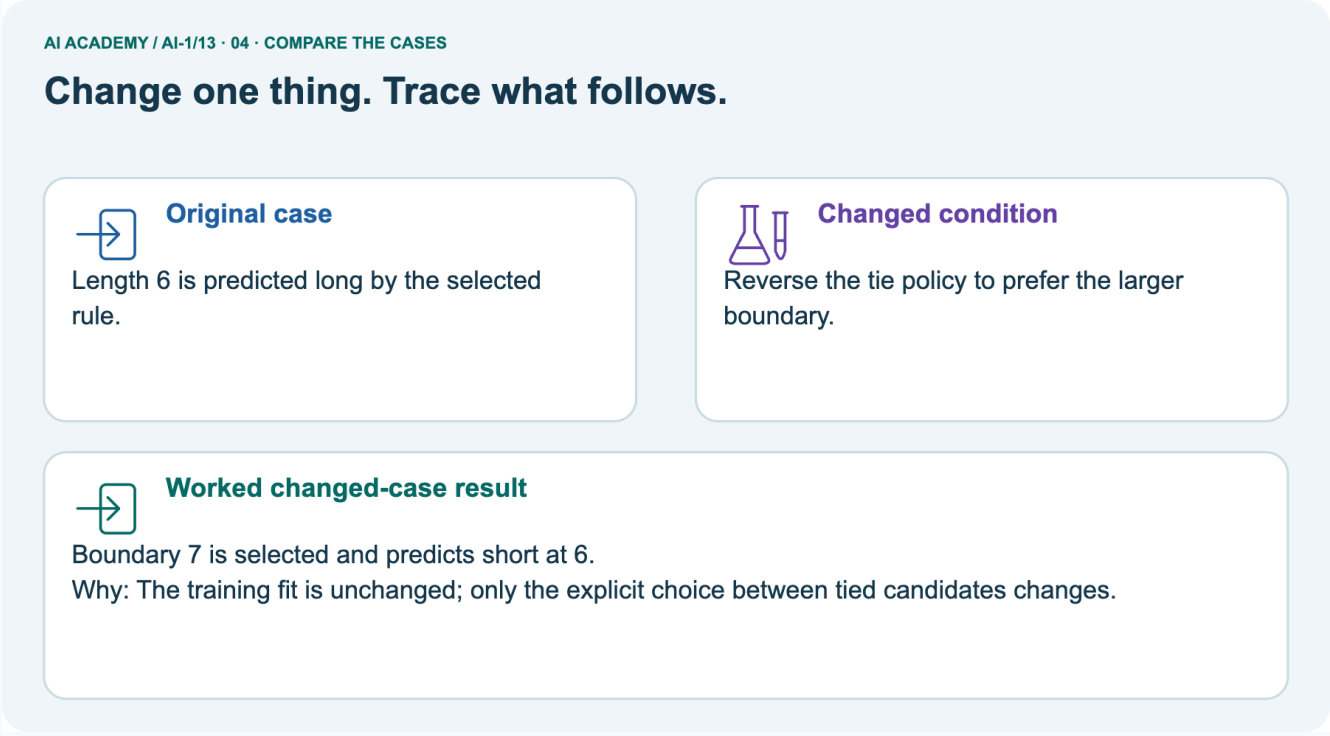
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Length 6 is predicted long by the selected rule. Changed condition: Reverse the tie policy to prefer the larger boundary. Expected result: Boundary 7 is selected and predicts short at 6.

Reason: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

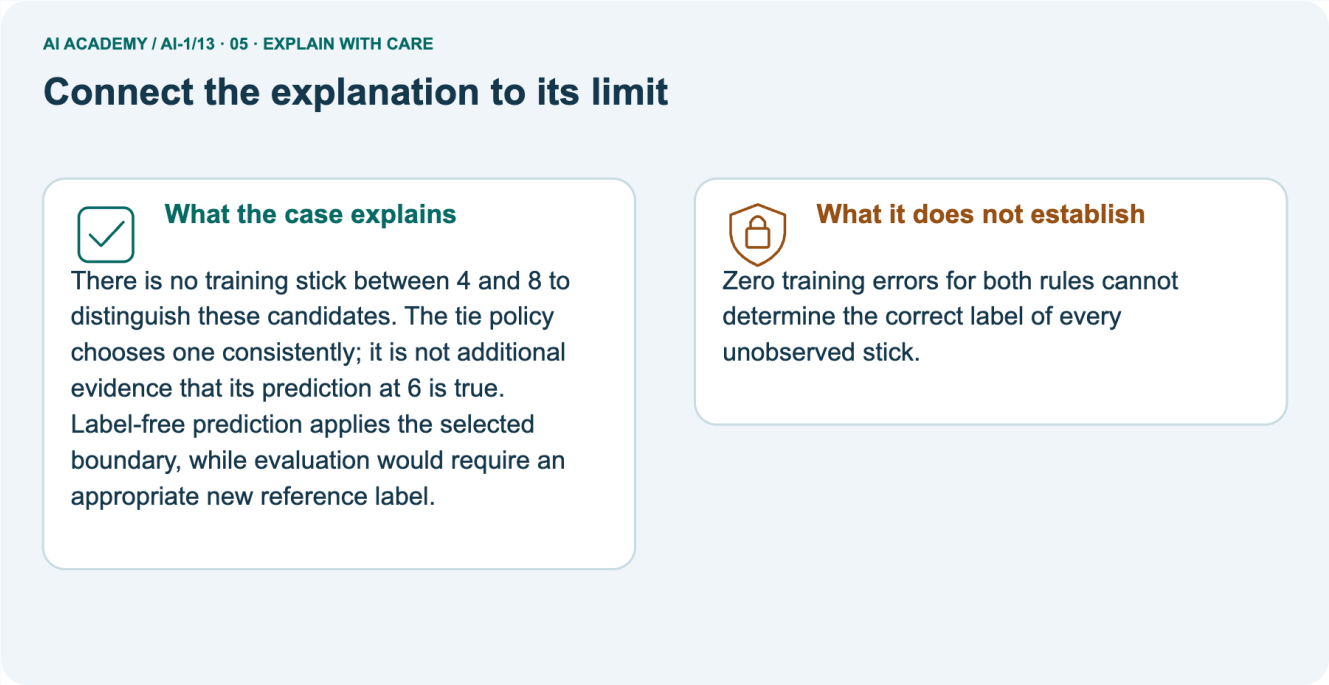
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

There is no **training** stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true.

Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label. Limit: Zero **training** errors for both rules cannot determine the correct label of every unobserved stick.

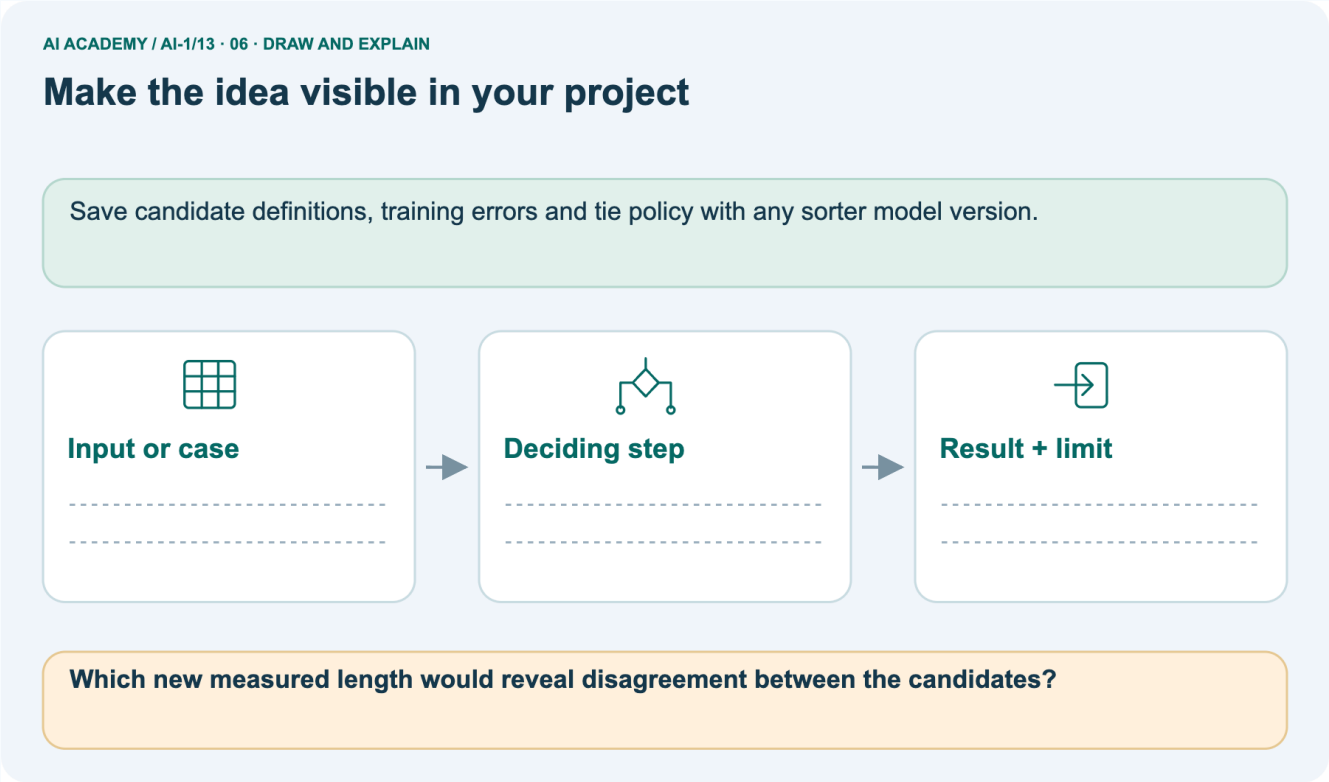
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Zero **training** errors for both rules cannot determine the correct label of every unobserved stick.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save candidate definitions, **training** errors and tie policy with any sorter model version. Draw your **input**, deciding step, result and limit.

Supported question: Which new measured length would reveal disagreement between the candidates?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Length 6 does: boundary 5 predicts long and boundary 7 predicts short. A checked label would then provide new **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.